AN IDEAL GENERATED BY A COUNTABLE COLLECTION OF IDEMPOTENTS IS PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The idempotent replacement step (e_{n+1} -> e_n + (1-e_n)e_{n+1}) is asserted with its purpose stated (ascending chain) and is elementary to check; content verbatim.

Editorial note. The label lemma-ideal-generated-by-idempotents-projective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Idempotents e_n and the ideals (e_n); directed colimits of modules and Hom-colimit adjunction
- Mittag-Leffler exactness: Algebra, lemma-ML-exact-sequence (surjective transition systems give exact lim)

Source reference. Original lemma 0598: sources/stacks-chow/tags.txt, source lines 6806–6806. Prerequisite pointer invoked at 2680; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let $I \subset R$ be an ideal generated by a countable collection of idempotents. Then I is projective as an R-module.*

Proof. Say $I = (e_1, e_2, e_3, \ldots)$ with e_n an idempotent of R. After inductively replacing e_{n+1} by $e_n + (1 - e_n)e_{n+1}$ we may assume that $(e_1) \subset (e_2) \subset (e_3) \subset \ldots$ and hence $I = \bigcup_{n \geq 1}(e_n) = \operatorname{colim}_n e_n R$. In this case

$$\operatorname{Hom}_R(I, M) = \operatorname{Hom}_R(\operatorname{colim}_n e_n R, M) = \lim_n \operatorname{Hom}_R(e_n R, M) = \lim_n e_n M$$

Note that the transition maps $e_{n+1}M \to e_n M$ are given by multiplication by e_n and are surjective. Hence by Algebra, Lemma 0598 the functor $\operatorname{Hom}_R(I, M)$ is exact, i.e., I is a projective R-module. □

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